

MUSTASAARI VALASSAARET, FinlandWMO: **029100**Lat: **63.436N**Lon: **21.064E**Elev: **11**StdP: **101.19**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.0	-15.1	-20.7	0.6	-17.9	-17.3	0.8	-15.0	16.4	-4.8	15.1	-4.8	4.2	140	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	22.4	19.0	20.8	18.0	19.4	16.9	19.7	21.9	18.4	20.4	17.3	19.1	3.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.5	21.3	17.5	12.5	19.9	16.4	11.7	18.6	56.1	22.1	52.0	20.4	48.6	19.2	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	13.0	11.5	DB	-18.5	23.8	5.2	2.2	-22.2	25.4	-25.3	26.7	-28.2	28.0	-32.0	29.6	(4)
(5)			WB	-18.8	20.1	4.9	1.5	-22.3	21.2	-25.2	22.1	-28.0	22.9	-31.5	24.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-3.8	-5.2	-2.4	1.9	6.6	11.6	15.9	15.8	11.5	6.1	1.9	-0.5	(6)	
	DBStd	7.89	4.58	5.64	3.75	2.37	2.73	2.45	2.63	2.42	2.58	2.97	3.08	3.58	(7)	
	HDD10.0	2310	426	427	385	244	110	11	0	0	14	124	242	327	(8)	
	HDD18.3	4878	684	660	643	494	363	201	85	84	204	381	492	586	(9)	
	CDD10.0	491	0	0	0	0	5	59	184	181	59	3	0	0	(10)	
	CDD18.3	17	0	0	0	0	0	0	11	7	0	0	0	0	(11)	
	CDH23.3	26	0	0	0	0	0	0	17	9	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0	(13)	
(14) Wind	WSAvg	5.7	6.3	5.9	5.8	5.4	5.2	4.9	4.5	4.9	5.5	6.5	6.6	6.6	(14)	
(15) Precipitation	PrecAvg	515	38	29	31	24	32	40	60	55	50	55	55	45	(15)	
	PrecMax	682	75	59	52	62	73	96	126	99	105	107	112	87	(16)	
	PrecMin	386	6	0	8	2	4	13	6	16	15	13	11	11	(17)	
	PrecStd	90	19	18	12	15	19	21	30	27	23	23	24	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	3.7	4.8	9.9	16.1	20.3	25.3	24.5	18.3	12.7	8.8	5.7	(19)	
		MCWB	3.0	2.3	2.7	6.3	11.4	15.5	20.9	20.6	15.7	11.5	7.8	5.1	(20)	
	2%	DB	2.6	2.5	3.5	7.7	13.6	18.0	23.1	22.0	17.0	11.7	7.7	5.1	(21)	
		MCWB	1.8	1.5	2.0	5.2	10.0	14.3	19.6	18.9	14.9	10.8	7.0	4.3	(22)	
	5%	DB	2.1	1.6	2.7	6.4	11.9	16.6	21.5	20.5	15.9	10.8	6.8	4.3	(23)	
		MCWB	1.4	1.0	1.5	4.4	8.9	13.6	18.6	17.9	14.1	10.0	6.2	3.7	(24)	
	10%	DB	1.4	1.0	1.9	4.9	10.5	15.4	20.0	19.4	14.9	9.9	6.0	3.6	(25)	
		MCWB	0.8	0.5	1.0	3.4	8.1	12.9	17.6	17.0	13.5	8.9	5.3	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	2.6	3.2	7.0	12.3	16.6	21.6	21.4	16.5	12.0	8.1	5.6	(27)	
		MCDB	3.4	3.5	4.5	9.2	15.0	19.1	24.2	23.8	17.8	12.5	8.6	5.7	(28)	
	2%	WB	2.1	1.8	2.2	5.5	10.4	15.0	20.2	19.2	15.4	11.0	7.2	4.6	(29)	
		MCDB	2.4	2.4	3.1	7.4	13.0	17.2	22.6	21.4	16.6	11.5	7.6	4.9	(30)	
	5%	WB	1.5	1.1	1.7	4.5	9.2	14.1	19.0	18.2	14.6	10.1	6.4	3.8	(31)	
		MCDB	1.9	1.5	2.4	6.1	11.5	16.1	21.1	20.1	15.6	10.7	6.7	4.2	(32)	
	10%	WB	0.9	0.5	1.1	3.5	8.3	13.3	17.8	17.3	13.7	9.1	5.5	3.0	(33)	
		MCDB	1.4	0.9	1.8	4.8	10.2	15.1	19.8	19.1	14.7	9.8	5.9	3.5	(34)	
(35) Mean Daily Temperature Range	MDBR		3.7	4.0	4.2	4.0	4.8	4.8	4.5	4.3	3.6	2.7	2.6	3.0	(35)	
	5% DB	MCDBR	3.2	3.2	3.9	6.1	7.1	6.5	6.2	5.7	4.6	3.0	2.3	2.8	(36)	
		MCWBR	3.1	3.0	3.0	3.9	4.3	3.9	3.5	3.5	3.3	2.6	2.4	2.8	(37)	
	5% WB	MCDBR	2.9	3.1	3.6	5.8	6.6	5.9	5.7	5.2	4.3	2.8	2.3	2.8	(38)	
		MCWBR	3.0	2.9	2.9	3.9	4.4	3.8	3.6	3.5	3.3	2.6	2.5	2.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.233	0.289	0.312	0.351	0.352	0.351	0.371	0.360	0.333	0.316	0.372	0.350	(40)	
	taud		2.040	2.154	2.214	2.279	2.448	2.498	2.466	2.492	2.542	2.384	2.645	2.570	(41)	
	Ebn at Noon		371	628	776	825	863	871	839	817	771	610	343	196	(42)	
	Edh at Noon		30	63	89	104	97	94	95	85	67	52	27	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.13	0.71	2.23	3.85	5.23	5.92	5.40	4.10	2.39	0.90	0.21	0.07	(44)	
	RadStd		0.01	0.14	0.26	0.37	0.48	0.46	0.48	0.36	0.23	0.17	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page